

A Packing of 79 Lines in $\mathbb{R}^{14}$

Limin Ge*

October 8, 2026

Abstract

We give 79 1-dimensional subspaces of $\mathbb{R}^{14}$ whose smallest squared chordal distance is at least 0.906830. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.906261, obtained by deleting 1 of the 80 lines of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 79 subspaces of dimension 1 in $\mathbb{R}^{14}$ whose pairwise squared chordal distances $D(a, b) = 1 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 0.906830 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 14$, $n = 1$. The cell $N = 79$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 0.906261, obtained by deleting 1 of the 80 lines of the packing found by this search. The Rankin simplex bound for this cell is $1(14 - 1)/14 \cdot 79/78 = 0.940476$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m14n1N79.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 78 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:
18506743310171703972313808798534688084957483089
denominator:
20408163265311101193540510318900463281045593250

B Coordinates

79 blocks of 14 rows with 1 integers each; divide by 10^{12} . File: `grassmann/packings/m14n1N79.txt`.

20210153780	-156952846863	80647535707	86940785869
405877197132	67884122485	-279632192739	278295625576
-168824076175	-99200943496	309259741223	-403398018862
186521248308	110153927034	-235139304592	168325734324
-441292540524	216483356313	182490437359	-28246327440
-97499763732	-523944514140	83866718781	626405691495
-11131933123	-373334837004	256383126696	253878865510
6347373579	248858636284	450515307	52032819765
686315447371	-323583685806	274313191456	122497976939
-117726611127	381247425006	30066393005	120720576063
-226564548749	-89294134694	74794668528	-341952167965
-38868143998	-167021465304	-648071247408	226027791892
58596314593	-303569803051	181475173696	-234393201805
-161213469384	219112248931	-351603892154	-105101185855
362943976711	-428218718162	575924058408	194395924423
165851347863	4616024310	153718173653	-17769093636
46107989957	-101422070657	143978898070	-85131141350
-180365583100	-157675458389	-110828757587	41630289644
-156959861617	203642858797	-126900943282	-423486276449
279448605792	330196012409	97191167884	155947619321
-153251092071	41712718219	205528904965	-657334106487
352282747039	383378715015	273926027716	-290957880329
207096941177	-112072763903	243569667049	190028289414
-474608239666	249016086620	352652952873	-71330983281
30560812521	177547882915	-53445721794	-27062994281
367477111226	-563228297787	186987159027	69340314136
-373837480215	-10923244359	111306108815	292985138247
109180497350	216771805691	-484707863487	316116360910
91919194810	-596578100697	390893231338	288192141350
-582111989901	-87736170003	52646195037	-185482143751
56078907481	497093558375	-384479472443	-232020653761
475110604718	-135837684979	48609821524	-14222232302
152133525665	30436889966	301779374594	-313345219657
355918000275	339588326475	-178550333818	48184551302
379509594028	387450931402	223025955741	119336133022
11423825860	-208877009266	-186050829287	31534220444
-124687328984	120927569212	-396008698066	-417983817118
209578539281	210362550728	458026443995	-130301141064
-162880750847	-21676045997	-127436271790	416143917965
58397550577	29465853959	297514556369	404165923985
109434007839	-17638741246	58104940352	426859496887
-168686054060	17985948833	-110429926466	48369076824
221324247901	361015635498	118753318008	-379777071392
380262622109	744875833072	118255042407	47593401109
-119861686518	90718923526	-139252281187	99389553966
-25414016180	131261281861	79114684709	-137983411837
-86486326420	-244985010801	397919732481	130635198901
-366490306462	131391328697	-398320981247	-143432953134
103298733258	-202429635166	76366082601	72615192553
7116730934	-1544625406	-397530132940	95817476461
57810001513	-254509903811	-108838416467	455488010864
107182647286	86258003235	-117956530194	38317882988
-182634163588	-292695781420	348385918643	70507099653
194042765600	-101006866828	-435591932298	226241940497
-704898374155	53748342989	-352503092909	605390466389
-236996302149	11355657297	-65347580463	-375369344621
76191012960	-137772308259	-179365846950	378621224043
-300358779030	-393405992078	-192637218595	-541987653291
112560904844	-585977506126	364535328027	191694457260
222287257218	246225541386	-382651309917	147557317292
302180023491	-85132543559	-70243986609	-90881063120
-383548418742	-114948317963	-78484544875	83210314582
-64909314613	32713540362	403393473641	-334603268214
159619483130	57369662443	231144462990	235275345346
118963026022	-462129569932	-215607210918	196328318840
-590112410543	47247788215	-540161909704	-283502807736
-217318924350	212576978626	-8389544995	193397708837
-15922019114	-145010301573	221857979774	40473194912
-399287361780	-215855718981	116046826080	161697767212
66991904440	-262345188965	152128545056	-371177443086

139364184017	207105233696	241065860277	-196168587283
322637521713	-27544086334	-292268812069	153665918114
94853108362	115598753301	4049282891	-450728053202
182841465167	77040054306	-170906746316	281034549680
98336415472	111269014675	23103929518	-195889089582
-376117143924	54932408399	-433987903211	7061635584
351231648744	-225410908135	371905159358	-399665201797
165541466530	-465464363761	95861781391	239463049176
314112057607	9032665736	55395264898	-355672789885
514906262157	-276028279266	-343872322640	-20344823600
242959469744	112836358067	-135288642629	-192177006146
-285158825871	305069840489	83928197271	43685809408
14922062124	-169197627447	-525758014568	-11812367025
166704421278	215042472167	-61725846566	-158490187643
71145340249	-433317315654	-347101517256	-478834818823
-232427434506	-107825501989	271451955297	-172351822492
522760177919	705415614043	270879019859	370253493596
-49439621849	-167364115298	299252091284	-22529113678
-254195177709	-173614832525	-223364223088	-412726109893
261396169644	15788743788	-136445122716	167338063594
-105799158943	30104932213	-4945123472	125116211686
348244224692	-32868669758	11560302026	379219398344
51802638225	59318894868	484082887249	345502060805
347201352130	-241702904444	198553250853	-63041117267
-58898185328	-422945264173	-223739722400	42960048130
-58878510505	-140932991344	212016051653	78989951932
-517975918836	253677901043	220118892946	449022886982
-36272310480	365240190461	249168520828	-369019091039
-175726465973	-326100849225	-410272239605	158803139788
180860064011	-446723953958	-89514629862	100760047512
51194778018	194271848915	484867051260	71373301223
435756240094	-85160931820	-276362584463	404105474585
62396008095	27312331322	101355579949	-199675938909
244718063358	7199942493	-149844070501	-96597454275
170947440333	91810459710	1876146303	120234023455
-285290962558	-212882587197	273586352072	-242393415249
-307501637653	-416327363975	220876654796	385709817561
343749480643	-157257424985	189191753851	456683641745
365986649137	-408814468104	97948737591	121029451956
242242785349	308484859034	-118291256870	-101452957976
-384720210362	-223699353127	-370796496051	-346839026942
-127094020433	150305226681	326305624784	391133874617
-223302128255	77757003198	-89444839590	203258439705
560099002121	30413786646	-486881872321	111006421712
111038096050	-383189163705	135543720811	-226664002075
-50896295911	173378844798	289270644092	-146073264475
179501345052	91765274319	466102331477	-308436984962
144147849415	132885392171	93056255358	210406951492
-231497566413	-428911800129	642530403622	294139916547
407313402177	-204277854145	-104668486685	134172656731
153066581385	450295601793	26743697637	82086116419
439616923167	163306937051	-76535103987	136990301682
63895434246	256133574454	2864433289	140574628247
304191928002	-332187772160	-64151682103	521593922670
-167100488499	-448572458955	-88032769384	133775023940
-88206366586	318714362454	-187630195066	294080023891
-141370694829	-269639563226	574194864994	215685265466
-181492143127	-121363746738	-19205509733	463827672269
346568774741	375083897213	46593485231	-261452229570
204872296326	170131851517	-283450386276	258397539973
394230757230	312445727674	-254408597260	193961245263
-57906404154	-104247752484	212697991608	447160126632
119236978696	13880345022	342467207169	322427390700
-331264359477	265086625031	70586263924	-394952376481
-184736295252	-156751871703	415285643617	-202990326884
-434100588997	246289853017	139010024689	-355756279135
175245049307	337833543806	-571121803447	284152080628
110440399573	7405476209	-357592742113	79477148459
170196343316	-92878150589	94373235318	34549837895
-455875263717	20809972551	242167146314	248254139425
61360017223	-123334539481	47370042387	68626980382
-98925616158	126110175887	-164613933746	212116479867
-95364361791	275572242832	-58744395946	35057289638
-387050052313	-297800806302	-53740581413	-56092786211
520316367556	183137307780	-21127322699	-140282197661
272274351072	265329937316	358073923525	399429884244
392582008411	-579672032221	-298989373576	48913393569
-200491303776	-187187963895	337362513012	575012370140
135433662121	435849564044	272340655617	-437595929303
-325382643448	126690935200	-305546660517	385822712963
-105711986563	-60373768149	-391625111360	-324939740048
-164639108071	-529342810195	-125081392813	61395007694
-214048471383	-191950584330	240506149256	164076070354
282935992477	184845620540	431845306303	79583634237
208426987544	150720335659	10001763424	8865575528
227922593597	-361960447497	151511288050	18216957556
230250708991	-294845987154	-209160920866	731169926
179332216322	95383648702	-289175519611	67321158824
271965903005	-219118967318	101824757057	-300525387203
-574100123623	22177709766	-234776014487	546093394016
-88777820161	-274318679772	398076006081	290745616924
115254136620	309478731135	227345090710	-105624501896
-3907655807	419145522598	363041159854	-23613793592
29951636252	12582550767	-256984366948	237892631447
-364214604029	-303105931430	352028940066	135796230766
-177219643652	-75338325032	-118872658075	-41729264915
289550930658	-219876251996	-42915448394	78336329481
-381970464918	93872198078	443301923132	445530881757
177682851280	59247359694	-150245579731	-22493575726
429815847143	-312337709292	-131163144322	-478300641033
-491377743712	-584741535430	97359677143	-173474269685
-346622047884	-201660620835	-379795602722	-174902842263

-130186670743	230943736102	461122626041	329301134639
-292132656618	162237583528	-234732322674	-113724429323
-326261274213	179622562858	48249973187	199589882799
-267509275246	-103784931583	-291592829680	180032045993
298033068921	-481853282848	89806427179	-74195113048
245627851937	35494297052	-416859252395	-110400251703
-63548562311	-75641706482	444565004457	41037235679
484860313259	-605523800579	-297501937742	420218233878
138327868546	132143302863	-301544154404	-578263634251
474723918200	405825076183	1419703381	-114728453084
-124546929439	-220139337060	49315670803	-232655153716
-103556523247	-61533312018	-239739320482	324349126999
66506057969	255405377127	-308119108120	-178886562119
-119244567126	311204859544	201933612266	-267463420568
248327683824	148959091569	-113179417151	134654324381
-128749608415	14749562567	121785353341	117995029
461361276439	-551912737655	283955336214	-62713101006
225643336451	-428684348432	-370396273100	703601904832
137491671911	197794757156	248396758454	125015387415
-24812508406	19149415873	123192439084	-267044486100
37546396197	-135304190573	140581658281	191981387861
55533211195	-186909380088	-130149221778	81739238153
-189662243944	-90671884335	299980214057	-386621314890
736082392269	-27656992072	-276740549883	-80878870902
-177971918958	219472048121	-63163105950	140851266599
-76609958928	-220265231230	-225596561697	-83297831340
-169410668648	126783508574	-381347675139	169167094922
-118280007781	-265079435458	-222587251493	375482779363
158137955702	90121007270	-464326992728	-301874800616
76614111182	75433549658	106141900025	-86422355299
-42810541718	257653771546	115168207023	23655982101
262129736043	96619080424	479440605813	-522092516821
-309537815857	147011218735	-29376375696	44517914527
-307452054675	250727408397	-167100016429	-79810746101
-318894625011	-506464240591	191328307436	-173761315012
531686539062	614387902895	46992541848	-358088958204
-432386379487	-270980276820	-337794958189	-84217081493
121906628616	350319326931	182830867144	75255589710
-54604148123	449754665580	129433997524	135797543898
276337087146	314648555605	-345717646447	485387586729
71856805734	124293708477	282184213396	305416284705
478472709776	217785585095	-242763181903	318454350446
429192614455	75893745323	-516926888909	-28165649061
-129476738976	224652514490	128119935736	455089176538
69157982240	-249773193223	-386576950444	152895637326
-410676226987	-263105512896	277419582049	312404705011
220311900120	-199184819435	-459653331383	509723324436
-60511454336	77749853484	254443580531	286263348726
-341945703229	-333603882513	-48449512731	322395720492
101145415130	307949094336	120891081568	-37238632545
256609286982	-87316819777	-331753872412	-74433703351
375559624512	125643140079	-206810998410	170615959623
60534891988	390438593802	269116960686	-65811888101
-39725181372	358954043724	133504065806	16987904308
-219184028292	-63890862223	96740210383	425115562435
-350868734584	428087334140	-115544780432	-63586606576
318602230874	-361170832855	486693384590	167812360020
152193258414	-212497479899	165096760656	-108296831464
-61208375624	239224670945	193013338890	180327442974
-37246743569	6759586588	373959584429	-502987716544
-174439501868	-196216227739	155793521337	-154604733039
213192271036	220400243516	-81683517079	97143563361
-362904471335	-58093355625	-90719421676	-62188439094
-103935390896	-429689268387	110839446803	375945738600
-452528862259	-294549005724	-368978037331	288580163717
-521740316935	-339090966741	-399420325710	130789602959
54504421141	-272196882297	58752493601	552230372695
-26404011088	-117621328704	-540120108235	42451393593
-45300105881	107549098937	-98604960405	221916597512
102442791523	82423625252	335395235871	199153634154
-402842068564	163453840521	296269851244	-24515667366
-345453146469	362562856303	82768549538	-73620481724
-460981829963	-240909990840	-80647881515	145263320061
-291270801949	-4544419494	366516533133	-550965565257
-73626175479	-486349557634	42763330230	404801467333
-125319366506	182504894852	154121997030	-134295362296
-410094522484	-339793751400	-303944981440	110631429295
36688322544	300733830137	68893565903	177872823151
-415796939213	-169618166984	-304054341427	23763465520
24334408375	-261709296159	312239710119	544857894934
202538685463	118273106988	132797985950	-329833758812
-57795793268	262422883486	160222388001	-49008951188
-94284679234	18209675808	-183673249499	144881028324
89929505913	-222147413589	-480037536346	117980793929
168698317234	69346868972	-241031470149	308974558615
575876568035	854141605297	-101691520316	-234907247169
-384729534817	109615768428	228909599582	-364858932957
414744448688	62891194210	508950681381	-175293900631
98355975951	93596745515	-292357188339	373756324273
-27128755506	1444027730	-27521293885	86653354235
210899162842	78206993446	-667216192745	-35113786691
256535003183	70611610371	-285909168004	424076261330
27159849344	-403865445318	-110098454788	134876001384
-223718195411	-11122479350	48705477630	273405909295
61159955982	-113482814047	-227194468871	230643003234
357314715600	-486116525461	408177442075	82122522240
-81188031428	-173231185976	113597372060	-446760248015
-387261278893	-24150496891	-188292653017	-68918196673
-169167323692	216560959369	149903447303	-531168236567
-33779887865	-276763485239	238174662831	304893740741
-191786238563	-215596586547	157278168832	-135469011704
-219958604937	-178669345502	-42839075146	-94452595050

95450831318
48708975666
210872799868

-150542069896
225557199510
-493160132730

-86956347241
345039658579
-246738631551

-193485389757